

ChagolTalk Developer Blueprint v1

A concise software engineering guide for building ChagolTalk, an anonymous voice and text communication platform using ASP.NET Core MVC, .NET 10, SQL Server, SignalR and WebRTC.

Chapter 1 - Introduction

- Project vision
- Objectives
- Scope
- Target users
- Future goals

Chapter 2 - Functional Requirements

- User registration/login/profile
- Random matchmaking
- Real-time text chat
- Voice chat
- Friends
- Blocking
- Reporting
- Admin dashboard

Chapter 3 - Technology Stack

- ASP.NET Core MVC
- .NET 10
- Entity Framework Core
- SQL Server
- SignalR
- WebRTC
- Bootstrap
- Git/GitHub

Chapter 4 - Project Architecture

- MVC request flow
- Controllers
- Services
- Repositories
- Database
- Dependency Injection

- Middleware

Chapter 5 - Database Design

- ER diagram
- Users
- Friends
- FriendRequests
- Messages
- Calls
- Reports
- Notifications
- Relationships
- Indexes

Chapter 6 - Frontend Design

- Landing
- Login
- Register
- Dashboard
- Match screen
- Chat
- Profile
- Settings
- Responsive UI

Chapter 7 - Backend Design

- Controllers
- Models
- ViewModels
- Services
- Repositories
- SignalR Hub
- Authentication

Chapter 8 - Development Plan

- Create MVC project
- Configure Git
- Install NuGet packages
- Configure EF Core

- Identity
- Landing page
- Dashboard
- SignalR
- Matchmaking
- WebRTC
- Friends
- Deployment

Chapter 9 - Learning Resources

- Microsoft Learn
- ASP.NET Core Docs
- EF Core Docs
- SignalR Docs
- WebRTC.org
- MDN
- Bootstrap
- Git Documentation

Chapter 10 - Project Rules

- Naming conventions
- Folder structure
- Coding standards
- Git commits
- Testing checklist
- Documentation

Implementation Philosophy

Build feature by feature. Every milestone should compile, run, and be tested before moving to the next. Prefer clean architecture, meaningful commits, and documentation alongside code.

Project Milestones

- ☐ Authentication
- ☐ Database
- ☐ Landing Page
- ☐ Chat
- ☐ Random Match
- ☐ Voice
- ☐ Friends

- ☐ Admin
- ☐ Deployment